DIVIDING BY A REGULAR ELEMENT LOWERS THE DEPTH BY AT MOST ONE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The behaviour of depth along a short exact sequence. Invoked inside the counted proof (source lines 2051-2054). Gives the inequality for the quotient.

Source reference. Original lemma lemma-depth-in-ses: sources/stacks-dualizing/pinned-original-source.tex, source lines 1999–1999. Navigation pointer only: the counted text refers to the Stacks item lemma-depth-in-ses, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring, let $I \subset A$ be an ideal, and let M a finite A-module with $IM \neq M$.*

(1) *If $x \in I$ is a nonzerodivisor on M, then $\mathrm{depth}_I(M/xM) = \mathrm{depth}_I(M) - 1$.*
(2) *Any M-regular sequence $x_1, \ldots, x_r$ in I can be extended to an M-regular sequence in I of length $\mathrm{depth}_I(M)$.*

Proof. Part (2) is a formal consequence of part (1). Let $x \in I$ be as in (1). By the short exact sequence $0 \to M \to M \to M/xM \to 0$ and Lemma lemma-depth-in-ses we see that $\mathrm{depth}_I(M/xM) \geq \mathrm{depth}_I(M) - 1$. On the other hand, if $x_1, \ldots, x_r \in I$ is a regular sequence for M/xM, then $x, x_1, \ldots, x_r$ is a regular sequence for M. Hence (1) holds. □

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